

ROEBOURNE, Australia

WMO: 943090

Lat: 20.777S

Lon: 117.146E

Elev: 43

StdP: 14.67

Time Zone: 8.00 (E08)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	55.3	57.4	23.5	17.8	72.6	27.1	21.1	72.2	26.3	64.6	20.6	66.4	1.4	270	0.333

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.3	109.0	74.8	106.5	74.4	104.1	74.5	83.2	95.7	81.9	93.9	80.9	92.0	5.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.3	157.6	87.9	79.2	151.7	86.6	78.4	147.5	86.0	47.2	95.8	45.8	94.2	44.7	92.2	97.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.6	15.8	12.0	49.6	114.0	1.2	2.5	48.8	115.8	48.1	117.2	47.4	118.6	46.6	120.4		
(5)			42.6	89.9	1.3	5.3	41.6	93.8	40.8	96.9	40.1	99.9	39.1	103.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	81.6	90.5	89.3	89.0	84.3	77.1	71.2	70.0	72.6	76.9	83.2	86.3	89.0	(6)
(7)		DBStd	8.49	4.37	4.58	4.13	3.58	4.02	4.30	3.25	3.44	4.40	5.41	4.71	4.54	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	5	0	0	0	0	0	2	3	0	0	0	0	0	(9)
(10)		CDD50	11522	1254	1099	1210	1030	840	635	621	700	807	1028	1090	1208	(10)
(11)		CDD65	6051	789	679	745	580	375	187	158	235	357	563	640	743	(11)
(12)		CDH74	77625	11645	9827	10884	7266	3348	1203	1021	2093	3852	7240	8627	10619	(12)
(13)		CDH80	44176	7324	5915	6639	3914	1345	292	204	701	1854	4232	5209	6547	(13)
(14)	Wind	WSAvg	4.0	3.8	4.0	3.6	3.2	3.9	4.3	4.1	3.9	4.0	4.2	4.5	4.5	(14)
(15)	Precipitation	PrecAvg	11.7	2.3	3.1	2.0	0.6	1.0	1.4	0.6	0.2	0.0	0.0	0.4		(15)
(16)		PrecMax	25.8	12.9	11.9	12.7	4.5	5.8	12.2	3.3	1.1	0.3	0.2	0.5	3.9	(16)
(17)		PrecMin	2.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	6.1	3.4	3.1	2.8	1.1	1.4	2.4	0.9	0.4	0.1	0.0	0.1	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	111.5	110.8	108.0	102.6	96.6	90.5	87.9	92.5	99.0	108.0	109.8	111.5	(19)
(20)			MCWB	75.6	76.3	74.4	71.5	68.1	68.8	63.5	65.4	67.6	70.7	73.1	73.7	(20)
(21)		2%	DB	108.4	106.1	104.9	99.5	92.8	86.6	84.8	89.1	96.2	104.0	106.5	108.0	(21)
(22)			MCWB	76.1	76.9	73.9	71.7	68.0	65.9	62.5	64.3	67.0	70.2	71.9	74.4	(22)
(23)		5%	DB	105.1	102.8	102.7	97.4	89.9	83.5	82.2	86.5	93.2	100.6	103.0	104.2	(23)
(24)			MCWB	76.5	76.9	74.3	70.5	67.8	64.5	62.1	62.8	66.3	69.0	71.7	75.2	(24)
(25)		10%	DB	101.6	99.4	100.0	95.1	87.3	80.9	79.9	84.1	90.1	97.0	99.9	100.6	(25)
(26)			MCWB	77.1	78.0	74.3	70.6	67.4	63.5	61.6	62.0	65.9	68.6	71.0	75.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.9	84.6	83.5	80.8	78.2	75.5	73.2	73.6	75.4	80.1	81.3	84.9	(27)
(28)			MCD B	98.2	96.3	95.9	96.1	85.3	80.2	77.1	79.3	92.1	91.3	98.2	96.9	(28)
(29)		2%	WB	82.5	82.9	81.6	78.8	75.5	72.3	70.0	69.5	71.0	75.4	77.4	81.9	(29)
(30)			MCD B	96.5	95.0	91.5	88.1	83.2	77.9	77.1	80.2	85.9	89.5	93.5	94.5	(30)
(31)		5%	WB	81.3	81.7	80.5	77.4	73.7	70.2	67.1	67.1	69.2	73.5	75.9	80.4	(31)
(32)			MCD B	94.3	93.1	89.7	87.4	81.8	77.7	76.2	79.8	85.2	88.9	92.2	92.3	(32)
(33)		10%	WB	80.2	80.8	79.5	76.2	71.9	68.3	64.8	65.2	67.8	71.9	74.5	79.1	(33)
(34)			MCD B	92.4	91.1	88.6	86.5	81.0	76.9	74.6	78.8	83.4	88.1	90.5	91.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	20.3	17.9	20.0	21.2	20.0	18.8	20.9	24.0	25.7	25.9	24.0	20.9	(35)
(36)			MCDBR	26.2	24.0	24.4	24.7	23.0	21.4	24.1	27.2	28.9	29.9	28.5	25.9	(36)
(37)			MCWBR	7.3	7.3	7.9	8.2	8.6	8.9	9.7	10.5	10.3	9.7	9.6	8.6	(37)
(38)		5% WB	MCDBR	20.6	18.5	19.1	17.8	18.3	15.9	19.3	22.8	26.5	25.7	23.2	18.8	(38)
(39)			MCWBR	7.3	7.3	7.5	7.5	9.0	8.4	10.5	11.1	11.5	10.1	9.5	8.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.402	0.393	0.380	0.353	0.333	0.322	0.316	0.315	0.335	0.356	0.361	0.378	(40)	
(41)		taud	2.494	2.532	2.556	2.584	2.572	2.590	2.583	2.541	2.487	2.457	2.483	2.511	(41)	
(42)		Ebn at Noon	299	298	294	290	284	282	287	298	304	306	310	307	(42)	
(43)		Edh at Noon	37	35	33	30	28	26	27	30	35	37	37	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2361	2254	2017	1765	1478	1338	1453	1794	2119	2395	2551	2540	(44)	
(45)		RadStd	144	155	147	92	112	95	92	50	80	59	66	109	(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (8 neighbors)	+0.83	N/A	N/A	N/A	+0.86	+1.01	N/A	N/A	N/A	+284	(47)

Nomenclature: See separate page